

COMPARATIVE ANALYSIS OF HARDWIRED AND MICROPROGRAMMED CONTROL UNITS USING GEM5

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INTRODUCTION

Control units are fundamental components of a CPU, responsible for generating control signals to coordinate instruction execution. There are two primary approaches: Hardwired Control Units, which use combinational logic to generate signals, and Microprogrammed Control Units, which store control sequences in control memory. This project evaluates both approaches using gem5.

BACKGROUND

Hardwired Control Units generate control signals directly using logic gates, offering fast execution but limited flexibility.

Microprogrammed Control Units use microinstructions stored in control memory, allowing easier modification and support for complex instruction sets. Modern processors implement microprogrammed techniques combined with advanced pipeline and out-of-order execution mechanisms.

PROBLEM STATEMENTS

01.

To analyze and compare the performance of Hardwired and Microprogrammed Control Units using a cycle-accurate simulation environment, and evaluate their impact on execution time, CPI, and overall CPU efficiency

02.

Modern processors require a balance between speed and flexibility in control unit design. This project aims to study this trade-off by simulating hardwired and microprogrammed control units and determining which architecture performs better under different workloads.

OBJECTIVES

01.

To implement and simulate hardwired and microprogrammed control units using a cycle-accurate simulator (gem5) for realistic CPU behavior analysis.

02.

To analyze key performance parameters such as execution time, CPI (Cycles Per Instruction), and instruction throughput for both control unit designs under different workloads.

03.

To understand how hardwired control units offer higher speed while microprogrammed control units provide flexibility, and determine their suitability for modern processor design.

ABSTRACT

This project presents a comparative analysis between Hardwired and Microprogrammed control units using the gem5 architectural simulator. Two CPU models were evaluated: TimingSimpleCPU representing a hardwired control structure and O3CPU representing a microprogrammed, out-of-order architecture. Performance metrics including CPI, IPC, execution time, and total cycles were analyzed using a benchmark program. The results demonstrate the performance trade-offs between simple in-order execution and advanced out-of-order microarchitectural designs.

METHODOLOGY



The gem5 simulator (v23.0.0.1) was installed and configured.

Two CPU models were selected:

- TimingSimpleCPU (Hardwired Control Unit)
- O3CPU (Microprogrammed Control Unit).



Multiple benchmark programs were executed on both CPU models to ensure fair comparison:

- Hello (baseline)
- Fibonacci (arithmetic workload)
- Matrix Multiplication (memory-intensive)
- Sorting (branch-heavy workload)

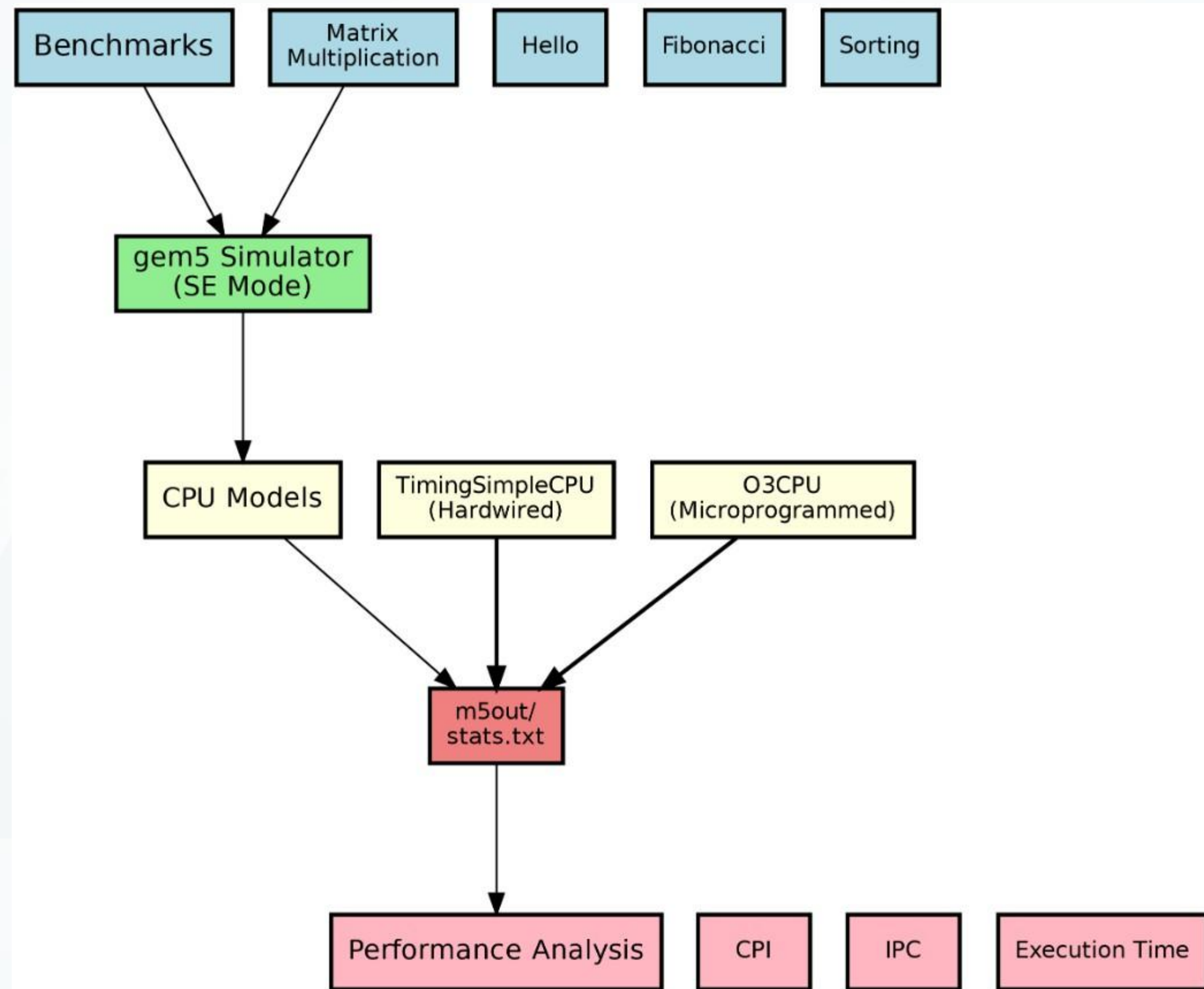


Performance metrics were extracted from stats.txt:

- Execution Time (simSeconds)
- CPI (Cycles Per Instruction)
- IPC (Instructions Per Cycle)
- Number of Cycles

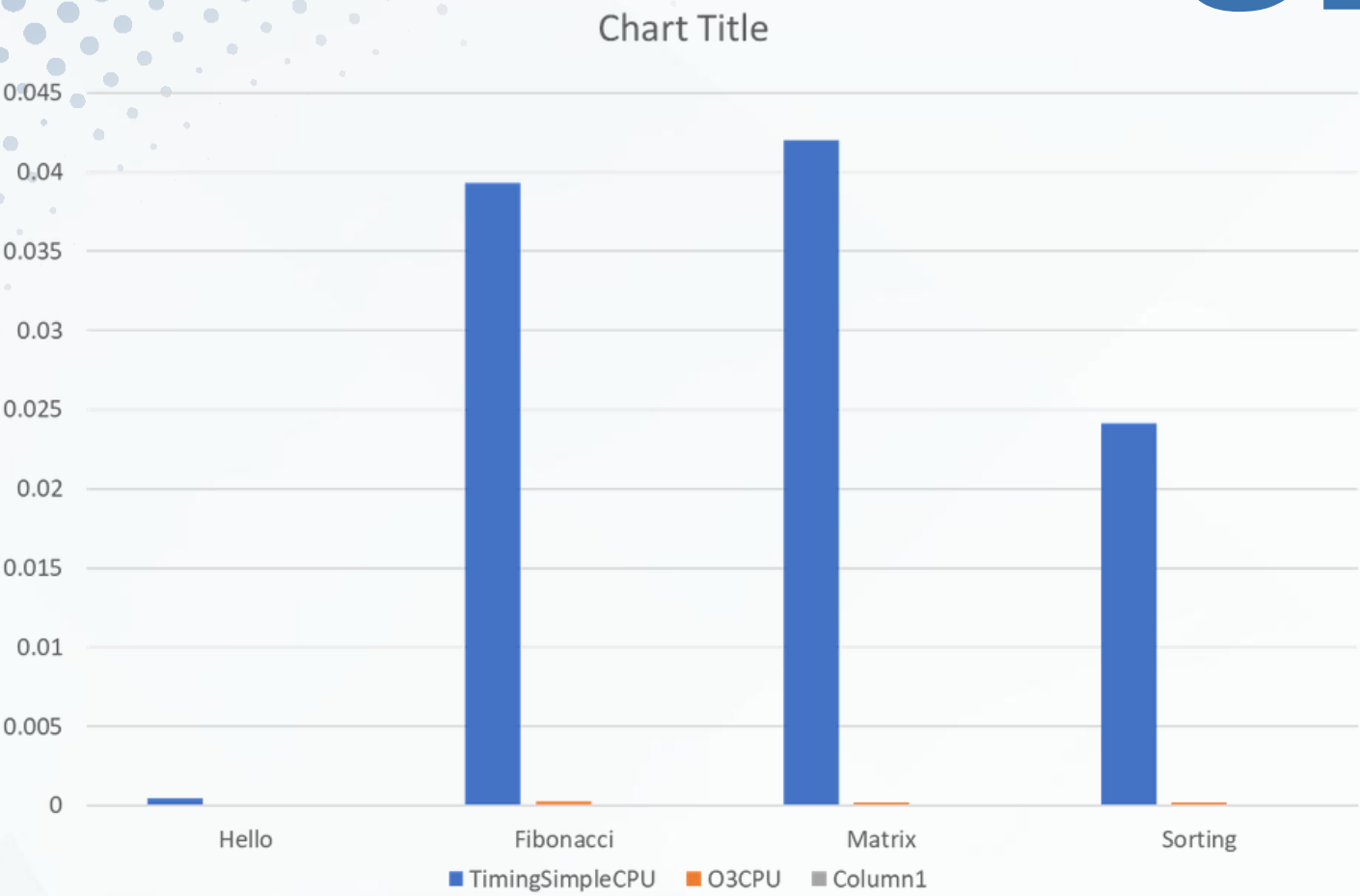
These metrics were analyzed to compare CPU performance.

SYSTEM ARCHITECTURE

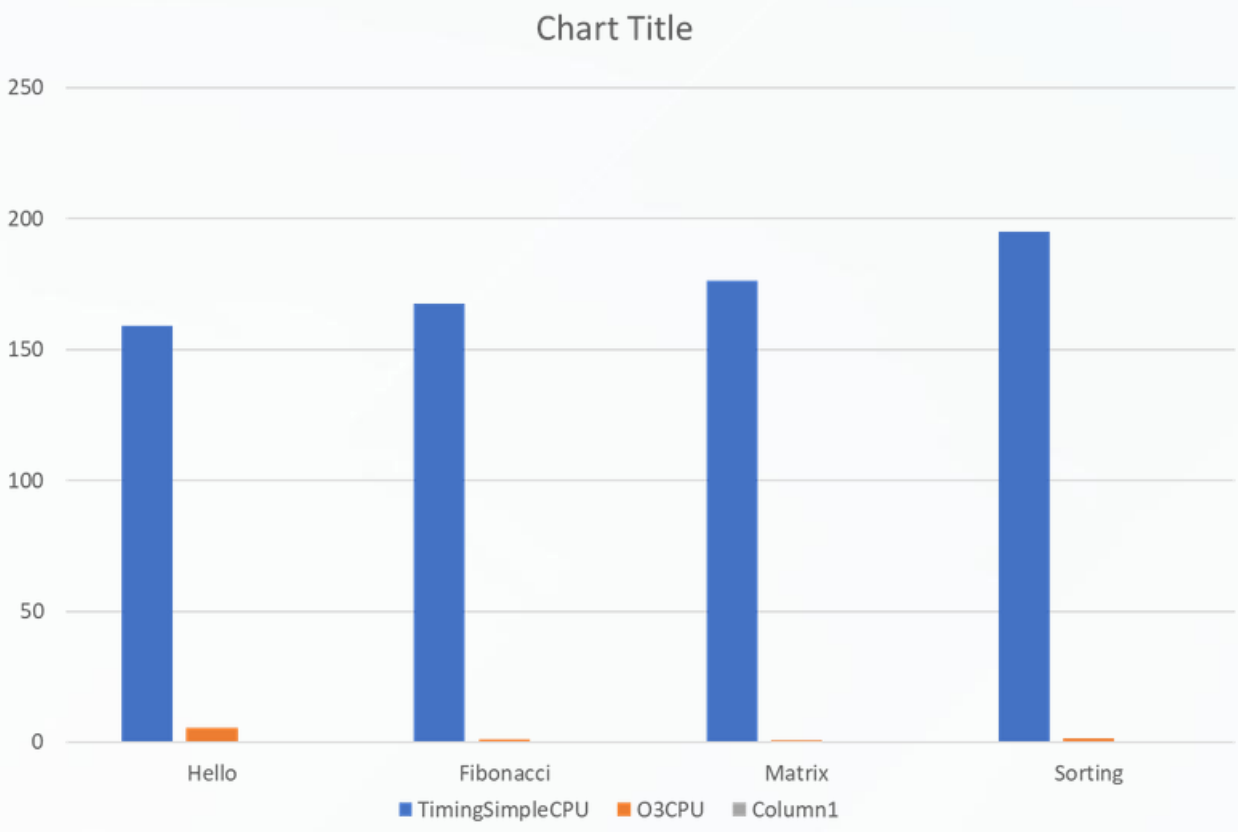


BENCHMARK DESCRIPTION

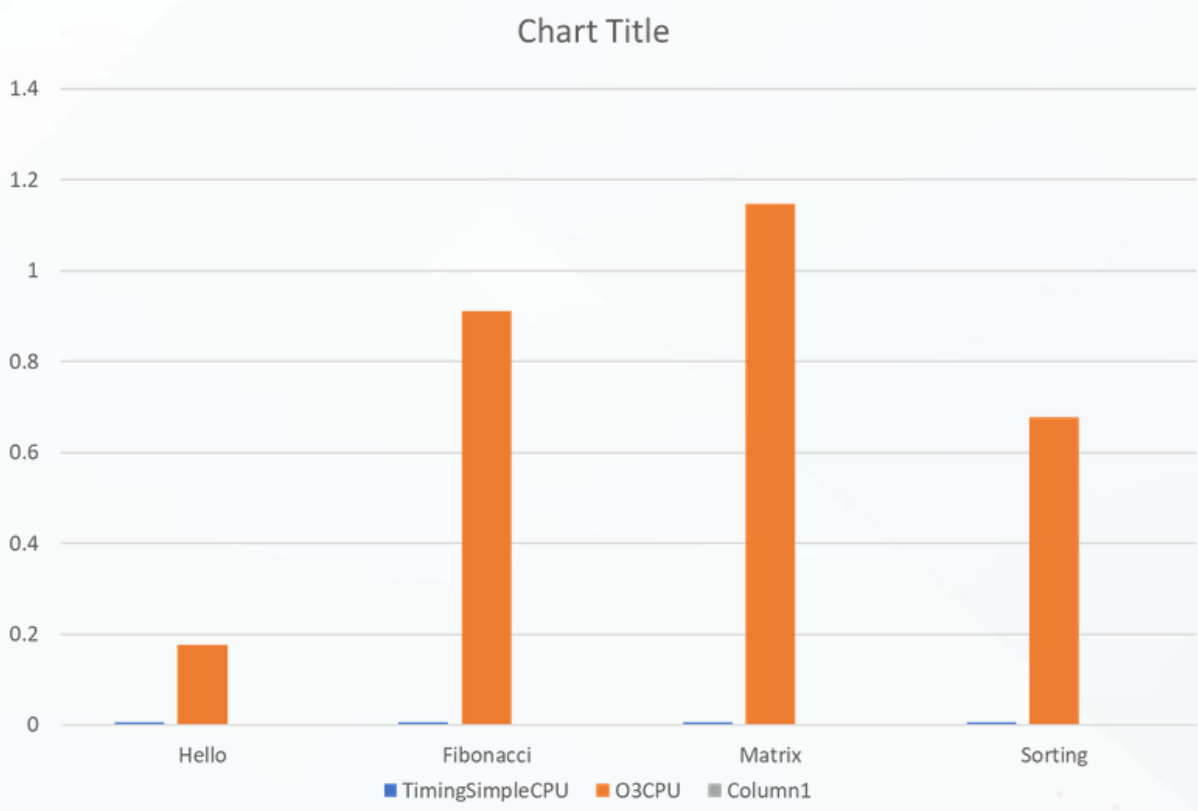
GRAPHS



GRAPH 1 — EXECUTION TIME



GRAPH 2 — CPI COMPARISON



GRAPH 3 — IPC COMPARISON

RESULTS

EXECUTION TIME

CYCLES PER INSTRUCTION

INSTRUCTIONS PER CYCLE

Key Observations

- O3CPU outperforms TimingSimpleCPU in all benchmarks
- Significant reduction in CPI and execution time observed
- Matrix multiplication shows highest speedup (memory-intensive workload)
- Sorting shows highest CPI due to branch-heavy behavior
- Higher IPC in O3CPU indicates better instruction-level parallelism

CONTROL UNIT COMPARISION

Key Insight

- Microprogrammed control units (O3CPU) provide significantly better performance due to instruction-level parallelism and efficient resource utilization
- Hardwired control units (TimingSimpleCPU) are simpler but suffer from higher CPI and lower efficiency

FUTURE SCOPE

- Extend current analysis by measuring power and energy consumption using gem5 statistics
- Implement and compare additional CPU models such as ARM-based configurations in gem5
- Modify cache parameters (size, associativity) to study their impact on matrix workload performance
- Analyze branch-heavy programs further to evaluate branch prediction behavior in sorting benchmark
- Scale benchmarks to larger inputs to observe performance trends under real-world workloads

CONCLUSION

- The project effectively modeled and compared hardwired and microprogrammed control unit architectures using gem5 under controlled conditions.
- Experimental results demonstrated that the microprogrammed (O3CPU) architecture delivers superior performance with reduced execution time and lower CPI.
- The study highlights a key design trade-off: hardwired control ensures simplicity, whereas microprogrammed control enables greater flexibility and enhanced performance.



THANK YOU